14. STRUCTURING THE VASCULAR SYSTEM : NOTES

DURATION : 03:24

STUDY SHEET

COURSE SUMMARY

In this video, we explore the structuring of the vascular system during embryonic development, focusing on the interconnection between the yolk sac and the amniotic cavity. The concepts of cephalisation, cardialisation, and diaphragmatic development are detailed, illustrating how the vascular system centralises as it develops. By analysing the cardinal systems and the formation of the venous vascular axis, we will discover how fluid movements and lateral germinations contribute to the formation of a 'protective aura', essential for the development of the cerebral system. This approach highlights the importance of reintegrating zone B when treating the nervous system, establishing a crucial link between the umbilicus and embryonic structures.

STRUCTURING OF THE VASCULAR SYSTEM

The **vascular system**, which develops at the periphery, is deeply integrated with the **vitelline vesicle**. **Wolff-Pander islands**, for example, form in the splitting of this vesicle.

Behind the vitelline vesicle is the **amniotic cavity**. Imagine it as a **balloon** containing this circulatory system, which is also found in the axis.

This "balloon" will deflate, and the entire peripheral system will centralise. This centralisation is orchestrated by the amniotic cavity.

Its differential growth rate, compared to the **primitive vascular field** and the development of the **brain**, generates a suction movement. Externally, this is accompanied by a **transverse thrust**, both thoracic and abdominal.

The longitudinal axis is more complex, integrating **cephalisation**, **cardialisation**, **diaphragmatisation**, and the entire **posterior peritoneum** down to the **perineum**.

This process is comparable to this integrating balloon. Simultaneously, at the periphery, there is a huge movement converging towards the **umbilical cord**. The cord, let us remember, is the meeting point between the **embryonic pedicle** and the vitelline vesicle, the latter gradually resorbing.

The secret here is **embryonic movement** and **developmental motility**: always bringing fluids towards the centre.

We will study the **cardinal systems** (subcardinal and supracardinal) in detail to finally arrive at a **venous vascular axis**. The latter presents specific vestiges.

The venous vascular system develops **transversely**, not longitudinally. It is composed of parts coming from different directions, resulting from a transverse union.

This contrasts with the arterial vascular system, which functions **longitudinally**, with two large conduits. On the venous side, these are **lateral germinations**, i.e., new trajectories.

When the fluids integrate towards the centre, **zone B** opens and takes shape, forming a kind of **protective aura**. This bubble contributes, among other things, to the construction of the **cerebral system**.

The first cavity to appear, or rather the second, is the amniotic cavity. It is at this moment that the **embryonic knob** becomes the future **epiblast**, the future nervous tissue.

The two must not be separated. If we want to treat the nervous system, we must always reintegrate zone B. This converges on the entire **median axis** and finally at the **umbilical** level.

The **umbilicus** is the final meeting point, the junction between the embryonic pedicle and the vitelline vesicle.